

Nawa Dahal

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Education

San José State University

B.S. Applied and Computational Mathematics

GPA: 3.93/4.0

San José, California

Aug. 2025 – Expected May 2027

De Anza College

Associate of Science - AS, Mathematics

GPA: 3.78/4.00

Cupertino, California

Sep. 2022 – Jun. 2025

Research Interests

Numerical Analysis, Scientific Computing, Nonlinear Partial Differential Equations, Applied Analysis, Dynamical Systems, and Mathematical Modeling.

Research Experience

A Matrix-Based Perspective on Multi-Level Richardson Extrapolation

San José State University

Undergraduate Researcher | Advisor: Dr. Mohammad Saleem

Sep. 2025 – Dec. 2025

- Developed a matrix-based formulation of multi-level Richardson extrapolation for numerical integration using a Vandermonde-type linear system.
- Established connections between Richardson extrapolation, polynomial interpolation, and the underlying linear algebraic structure of the extrapolation weights.
- Analyzed the effects of matrix conditioning, extrapolation weight growth, and finite-precision arithmetic on the stability and accuracy of higher-order extrapolation.
- Combined theoretical analysis with MATLAB experiments to investigate the tradeoff between truncation error reduction and finite-precision effects, demonstrating problem-dependent optimal extrapolation levels.
- First-author manuscript accepted for publication in *SIAM Undergraduate Research Online (SIURO)*.

Analysis and Numerical Investigation of Nonlinear Semiconductor Drift-Diffusion Systems

San José State University

Honors Undergraduate Researcher | Advisor: Dr. Daniel Brinkman

Jan. 2026 – May 2026

- Developed finite-difference discretizations and iterative solvers for coupled nonlinear drift-diffusion-Poisson systems modeling carrier transport in semiconductor devices.
- Analyzed equilibrium Poisson-Boltzmann equation using contraction mapping, spectral properties of the discrete Laplacian, and proved equivalence between quasi-linearization and Newton's method via explicit Jacobian derivation.
- Implemented Scharfetter-Gummel discretization and compared it with quasi-Fermi/Slotboom formulations, establishing asymptotic equivalence regimes and structure-preserving properties.
- Investigated physics-informed initialization, solver stiffness under varying doping and bias, current conservation, and recombination dynamics (SRH) through extensive MATLAB simulations.
- Simulated current-voltage characteristics of PN junctions in forward and reverse bias, comparing results with Shockley ideal diode law.
- Completed 70-page honors thesis integrating mathematical modeling, theoretical analysis, and numerical computation of nonlinear PDE systems.

Teaching Experience

MATH 19: Precalculus

San José State University

Instructional Student Assistant: Workshop Facilitator

Aug. 2025 – Dec. 2025

- Led weekly workshop sessions for approximately 25 undergraduate students, reinforcing precalculus concepts through guided problem solving.
- Explained mathematical concepts, answered students' questions, and guided group discussions during workshop activities.

MATH 33LA: Differential Equations and Linear Algebra

Instructional Student Assistant: Grader

San José State University
Jan. 2026 – May 2026

- Graded midterm and final examinations for approximately 160 students using detailed grading rubrics.
- Evaluated students’ solutions, identified mathematical errors, and provided written feedback on differential equations and linear algebra problems.

MATH 71W: Calculus for Business and Aviation

Instructional Student Assistant: Workshop Facilitator

San José State University
Aug. 2026 – Present

- Lead weekly workshop sessions supporting undergraduate students in calculus for business and aviation.
- Guide collaborative problem solving and reinforce core calculus concepts.

Publications

A Matrix-Based Perspective on Multi-Level Richardson Extrapolation 2026

Accepted for publication in *SIAM Undergraduate Research Online (SIURO)*.

Honors Thesis

Analysis and Numerical Investigation of Nonlinear Semiconductor Drift–Diffusion Systems 2026

Department of Mathematics and Statistics, San José State University.
Advisor: Dr. Daniel Brinkman.

Posters & Presentations

Numerical Investigation of the Poisson–Boltzmann Equation

Research Poster

- Presented at the **College of Science Student Research Day**, San José State University (Apr. 2026).
- Presented at the **Department of Mathematics and Statistics End-of-Year Research Showcase**, San José State University (May 2026).

Honors & Awards

2022–2025 **Dean’s List (Every Quarter)**, De Anza College

Cupertino,
California
San José,
California

Fall 2025 **President’s List**, San José State University

Relevant Coursework

In Progress Partial Differential Equations, Complex Analysis, Advanced Dynamical Systems

Completed Introduction to Analysis, Advanced Calculus, Ordinary Differential Equations and Dynamical Systems, Numerical Analysis I, Numerical Analysis II, Applied Probability and Statistics,

Planned Coursework Applied Mathematics I (Graduate Partial Differential Equations), Numerical Solution of Partial Differential Equations, Mathematical Modeling for Life Sciences

Technical Skills

MATLAB | C++ | LaTeX